Measurement Notebook

Student sheet

Name: _____ **Date:** _____

Safety: In corridors and on stairs, walk slowly and do not look at the screen; hold the box with both hands.

1. What I see on the screen

At the top: the **Quantity:** list, empty for the first few seconds; **Temperature** comes first and is chosen from the start. Beside it, the label field; **Enter saves** it, the same as the green **Save the value** button. The big blue number is the live value, **not yet saved**. Below: **Delete last**, **Clear all** (asks first, then cannot be undone!) and **Export data**. The chart shows only the points of the chosen quantity, and its axis does not start at zero: **read the numbers on the axis**. **One quantity per map:** every point remembers its quantity (the **Quantity** column), but a map compares only like with like. **Wait** 60 **s at every stop** (90 s for CO_2, 5 s for light), stand still, hold the box by its edges and do not breathe towards it. **Export before closing the window.**

2. What I do on my own

Protocol (circle one): **1** heat map / **2** light map / **3** CO_2 across spaces **Box:** _____
Quantity chosen: _____ **Unit:** _____ **Height:** _____ m **Start time:** _____
The exported file will be called: _____
First I predict. I do not change my prediction after I measure. The highest value will be at _____, the lowest at _____, and the difference between them will be _____, because _____

The labels. I copy them from here, letter by letter. Rules: **(1)** lower-case letters only; **(2)** no accented letters (`naive`, not `naïve`); **(3)** a hyphen, not a space; **(4)** at most three words; **(5)** only from the list; **(6) never leave the field empty** — the app writes `#1`, `#2` on its own, and those can no longer be grouped.

1. Heat map (8 stops, **Temperature**): `1 door` · `2 front-left-corner` · `3 teacher-desk` · `4 front-right-corner` · `5 centre` · `6 window` · `7 radiator` · `8 back-corner`

2. Light map (9 stops, **Illuminance**): `1 desk-1-window` · `2 desk-1-middle` · `3 desk-1-door` · `4 desk-3-window` · `5 desk-3-middle` · `6 desk-3-door` · `7 desk-5-window` · `8 desk-5-middle` · `9 desk-5-door`

3. CO_2 across spaces (7 stops, **CO2**): `1 outside` · `2 empty-room` · `3 occupied-room` · `4 corridor-break` · `5 library` · `6 sports-hall` · `7 outside-return`

Now I measure. Row n is stop n from the list above. **The circuit is walked twice**, in the same order.

#	Label	Time	Circ. 1	Circ. 2	Conditions
1					
2					
3					
4					
5					
6					
7					
8					
9					

The threshold: how big a difference has to be before I believe it.
Spread = the biggest Circ. 2 − Circ. 1 difference at the same place: _____ **Drift** = stop 1: _____ − _____ = _____
Threshold = spread + drift; e.g. $0.3 + 0.5 = 0.8$. Mine: _____ + _____ = _____
Highest: _____ Lowest: _____ Difference: _____, which ☐ **is** ☐ **is not** bigger than the threshold. *"Not detected" does not mean "does not exist".*

3. The conclusion I write

I predicted that, and the data show that _____

_____

The difference I trust is, because _____

_____

For someone else to repeat our measurement, we would need to give them _____

_____

Exit ticket

One thing that surprised you today, and one thing you would do differently if you repeated the measurement.

_____